

ALI RAZA

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AI Research Engineer · Medical AI · Generative Models · Computer Vision

RESEARCH INTERESTS

Medical Image Analysis · Vision-Language Models · Generative Models (GANs, Diffusion, Flow Matching) · Temporal Trajectory Modeling · Domain Adaptation · Explainable Clinical AI · MRI Reconstruction

EDUCATION

FAST-NUCES, Faisalabad, Pakistan

2022 – 2026

BSc Computer Science GPA: 2.94

Thesis: RadGuard — Multimodal Error Detection in AI-Generated Radiology Reports

PUBLICATIONS & ABSTRACTS

- Khan MM, **Raza A**, et al. “Generalizable Detection of CIEDs Using a Hybrid Transformer and CNN Framework.” *ACR 2026 — American College of Radiology Annual Meeting*, Washington DC. [Accepted]
- **Ali Raza** et al. “Segmentation & Classification of CIEDs in Chest X-rays.” *CIS-2025 Conference*. [Under Review]

RESEARCH EXPERIENCE

RadGuard: Multimodal Error Detection in AI-Generated Radiology Reports

2025 – 2026

Final Year Thesis · FAST-NUCES BioViL-T · CXR-BERT · MIMIC-CXR · FastAPI · Kubernetes · AWS EC2

- Designed multimodal VLM detecting 4 error classes (Supported, Hallucinated, Missing, Inaccurate) in AI-generated CXR reports — **Val F1=0.66** on 74,060-sample MIMIC-CXR dataset across 14 conditions.
- Proposed **ELRRS** (Error-Level Radiology Report Score): novel weighted reliability metric via multimodal fusion of ResNet-50 + CheXBERT aligned with ACR + RadGraph standards.
- Implemented **Bidirectional Cross-Attention** (BioViL-T + CXR-BERT); multi-task learning with pseudo-mask supervision; visual explainability via **GradCAM + CBAM**; FAISS-based RAG chatbot for diagnostic querying.

ProgressGAN-CXR: Temporally Conditioned GANs for Disease Progression Simulation

2025 – 2026

DCGAN · WGAN-GP · Spectral GAN · Knowledge Distillation · PyTorch · FastAPI · React · HuggingFace

- Introduced continuous severity conditioning into GAN-based CXR synthesis for smooth disease trajectory simulation (Normal → COVID-19); demonstrated **FID and Spearman r are orthogonal metrics** requiring separate evaluation.
- Knowledge distillation from WGAN-GP teacher resolved quality-semantics tradeoff: KD Generator achieves **FID 143** matched to DCGAN while improving **Spearman r to 0.984**; DCGAN augmentation achieved **perfect recall (1.00)** on rarest class.
- Deployed as full production application: React frontend (Vercel), FastAPI backend (HF Spaces), 4 live model architectures, Prometheus metrics. Live: progressgan-cxr.vercel.app

CIED Detection Pipeline — ACR 2026 Accepted Abstract

2024 – 2025

DINOv3 · DenseNet-121 · TransU-Net · nnU-Net · SAM · Vision Transformers

- Three-stage pipeline: DINOv3-based segmentation isolates device; hybrid dual-branch classifier identifies manufacturer (CLS + pooled patch tokens, 1536d); hierarchical model classifier conditioned on manufacturer embedding (64d prior).
- Handles class imbalance via FocalLoss, WeightedRandomSampler, Mixup, and SMOTE in hybrid feature space; generalizes across unseen device morphologies.

TECHNICAL SKILLS

Medical AI: BioViL-T, CXR-BERT, BioViT, CheXBERT, TransU-Net, U-Net, nnU-Net, SAM, DINOv3, ViT

Generative Models: GANs (DCGAN, WGAN-GP, Spectral), Diffusion, Flow Matching, Knowledge Distillation, Mamba SSM

CV & NLP: YOLO (v8/v11), ResNet, DenseNet, GradCAM, CBAM, BERT, Transformers, LoRA/QLoRA, RAG, FAISS

Stack: PyTorch, FastAPI, React, Docker, Kubernetes, AWS EC2, HuggingFace, Vast.ai, Git, Power BI

Datasets: MIMIC-CXR (PhysioNet approved), NIH ChestX-ray14, MS-CXR, CheXpert, FastMRI

INDUSTRY & APPLIED AI EXPERIENCE

Freelance AI / ML Engineer | Self-Employed

2023 – Present

Production AI Deployment · Voice AI Agents · Multimodal Systems · RAG Pipelines

- Deployed GPU-backed **YOLO multimodal detection system** (FastAPI + Vast.ai + React/Vercel) for automated asset depreciation detection in production.
- Built **Voice-Driven B2B Lead Gen Agent** (LiveKit STT/TTS, n8n, MCP) and **Voice-Cloned PDF RAG Assistant** (~80% cloning accuracy) — all production-deployed.

Data Analyst Intern | TAKMIL

2024

Louisville, Kentucky, USA · Remote

- EDA on 100K+ row educational datasets; **Power BI** dashboards; pipeline optimizations improving efficiency by **30%**.

SERVICE & ACTIVITIES

- **Co-Head Hackathon** — ACM Society, Daira 2025 (organized and led university-wide AI hackathon)
- **Member**, ACM Society Management Team — Daira 2024